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## Education

**Institute of Science Tokyo (formerly Tokyo Institute of Technology)**[1] **Tokyo, Japan**
*Doctor of Science in Mathematics* *Apr 2024–Mar 2026*
Supervisor: Kazuma Shimomoto

**Tokyo Institute of Technology** **Tokyo, Japan**
*Master of Science in Mathematics* *Apr 2022–Mar 2024*
Supervisor: Kazuma Shimomoto (2nd year), Fumiharu Kato (1st year)

**Tokyo Institute of Technology** **Tokyo, Japan**
*Bachelor of Science in Mathematics* *Apr 2018–Mar 2022*
Supervisor: Fumiharu Kato

## Professional Position

**JSPS Research Fellow (PD)** **Apr 2026–Mar 2027**
*Institute of Science Tokyo*
Host Researcher: Kazuma Shimomoto

**JSPS Research Fellow (DC1)** **Apr 2024–Mar 2026**
*Tokyo Institute of Technology*
Host Researcher: Kazuma Shimomoto

## Research Interests

Commutative algebra in mixed characteristic (via arithmetic methods such as perfectoid rings, prismatic cohomology, and almost mathematics).

## Papers and Preprints

16. **R. Ishizuka** and S. Yoshikawa, "*Hodge–Tate splitting and Akizuki–Nakano vanishing in positive characteristic*", arXiv:2609.00567, 2026.

15. M. Asgharzadeh and **R. Ishizuka**, "*Homological detection by perfectoid algebras*", arXiv:2607.17801, 2026.

14. **R. Ishizuka** and L. Navarro Chafloque, "*On perfectoidization of finite algebras over a perfectoid ring*", arXiv:2606.12229, 2026.

13. **R. Ishizuka** and L. Navarro Chafloque, "*Decomposition theorem for perfectoid rings along general ideals*", arXiv:2606.06241, 2026.

12. **R. Ishizuka** and S. Miyashita, "*Trace ideals of exterior powers of the module of differentials*", arXiv:2605.00418, 2026.

11. **R. Ishizuka** and S. Yoshikawa, "*A local-global correspondence for perfectoid purity*", arXiv:2604.25265, 2026.

10. **R. Ishizuka** and S. Yoshikawa, "*Algebraization of absolute perfectoidization via section rings*", arXiv:2604.02682, 2026.

9. **R. Ishizuka** and S. Yoshikawa, "*Derived graded modules*", arXiv:2601.19164, 2026.

[1] In Oct 2024, Tokyo Institute of Technology was merged with Tokyo Medical and Dental University and reorganized into *Institute of Science Tokyo*.

8. **R. Ishizuka** and S. Yoshikawa, "*Graded perfectoid rings*", arXiv:2511.02322, 2025.

7. **R. Ishizuka** and K. Shimomoto, "*Quasi-canonical lifting of projective varieties in positive characteristic*", arXiv:2506.01345, 2025.

6. **R. Ishizuka**, "*A higher algebraic approach to liftings of modules over derived quotients*", arXiv:2503.17964, 2025.

5. **R. Ishizuka**, "*Perfectoid towers generated from prisms*", Nagoya Math. J. **261**, e17 (2026). arXiv:2409.15785, 2024.

4. **R. Ishizuka** and K. Nakazato, "*Prismatic Kunz's theorem*", J. Algebra **693**, 732-769 (2026). arXiv:2402.06207, 2024.

3. D. Dine and **R. Ishizuka**, "*Tilting and untilting for ideals in perfectoid rings*", Math. Z. **307**, 66 (2024). arXiv:2308.09600, 2023.

2. **R. Ishizuka**, "*A calculation of the perfectoidization of semiperfectoid rings*", Nagoya Math. J. **255**, 742-759 (2024). arXiv:2305.07916, 2023.

1. K. Shimomoto and **R. Ishizuka**, "*A mixed characteristic analogue of the perfection of rings and its almost Cohen-Macaulay property*", arXiv:2303.13872, 2023.

## Talks

42. Dec 2026. *"Hodge–Tate splitting in positive and mixed characteristic"*, Algebraic Number Theory and Related Topics 2026, The University of Tokyo, Japan.

41. Nov 2026. *"Perfectoid criteria for Cohen–Macaulayness and Gorensteinness"*, The 47th Japan Symposium on Commutative Algebra, Okayama International Center, Japan.

40. Oct 2026. *"Hodge–Tate splitting and Akizuki–Nakano vanishing in positive characteristic I"*, Kinosaki Algebraic Geometry Symposium 2026, Kinosaki International Arts Center, Japan.

39. Sep 2026. *"Local and global aspects of perfectoid pure singularities"*, The 12th Japan-Vietnam Joint Seminar on Commutative Algebra, Khanh Hoa University, Vietnam.

38. July 2026. *"Decomposition of de Rham complexes in mixed characteristic"*, Honjo Waseda Algebraic Geometry Seminar 2026, Honjo seminar house, Japan.

37. June 2026. *"Graded modules and comodules: The dual notion of modules"*, The 37th Seminar on Commutative Algebra in Japan, Chiba University, Japan.

36. Dec 2025. *"Global singularities in mixed characteristic and graded perfectoid rings"*, Mini-workshop on Arithmetic Geometry in Sendai, Tohoku University, Japan.

35. Nov 2025. *"Perfectoid towers arising from Frobenius lifts"*, The 46th Japan Symposium on Commutative Algebra, Hotel Fukuracia Osaka Bay, Japan.

34. Nov 2025. *"Perfectoid towers arising from prisms"*, Workshop on Number theory at Tsuda University 2025, Tsuda University, Japan.

33. Oct 2025. *"Absolute perfectoidization of schemes and local to global principle in mixed characteristic"*, Singularity Theory Seminar, Nihon University, Japan.

32. Sep 2025. *"The Frobenius-Witt cotangent bundle"* (Survey talk), Yatsugatake Workshop, 2025, Wellness Garden in Goddess Forest : Kobuchizawa, Japan.

31. Sep 2025. *"A version of Kunz's theorem in mixed characteristic"*, Noda Algebraic Geometry Symposium, Tokyo University of Science, Japan.

30. July 2025. *"On the derived deformation functor of Frobenius liftings"* (Poster Session), 2025 Summer Research Institute in Algebraic Geometry., Corolado State University, USA.

29. July 2025. *"On computation of Tor modules over a derived quotient"*, The 36th Seminar on Commutative Algebra in Japan., Kyushu University, Japan.

28. June 2025. *"A unified construction of perfectoid towers by prisms"*, $p$-adic and Characteristic $p$ Methods in Algebraic Geometry., EPFL, Switzerland.

27. May 2025. *"Approximation of perfectoid rings by Noetherian rings and prisms"*, MPIM Algebra Seminar., MPIM, Germany.

26. Mar 2025. *"On the vanishing of Ext and liftings of modules on derived quotients using higher algebras"*, MSJ Spring Meeting 2025., Waseda University, Japan.

25. Mar 2025. *"Frobenius liftability and derived algebraic geometry"*, The 29th Conference on Algebra for Young Researchers in Japan., Osaka University, Japan.

24. Mar 2025. *"Liftability results from formal moduli problems"* (including survey), Small workshop on arithmetic geometry 2025 in Hakodate., Hakodate Community Design Center, Japan.

23. Nov 2024. *"Prismatic Kunz's theorem"*, The 45th Japan Symposium on Commutative Algebra., RIMS (Kyoto University), Japan.

22. Nov 2024. *"Applications of higher algebra to the lifting problem of modules"*, Tokyo Commutative Algebra Seminar., Online (Zoom), Japan.

21. Oct 2024. *"$F$-liftability obstruction in singular varieties through derived algebraic geometry"*, Singularity Theory Seminar., Nihon University, Japan.

20. Sep 2024. *"Frobenius maps on mixed characteristic rings via prismatic cohomology"* (Poster Session), L-functions and Motives in Niseko 2024., Setsu Niseko and Niseko Residents Center, Japan.

19. Sep 2024. *"A generalization of Kunz's theorem to mixed characteristic via p-adic cohomology theory"*, MSJ Autumn Meeting 2024., Osaka University, Japan.

18. July 2024. *"Perfectoid spaces, tilts and untilts"* (Survey talk), Atelier de Géométrie Arithmétique 2024., RIMS (Kyoto University), Japan.

17. July 2024. *"Prismatic approach to a mixed characteristic Kunz's theorem"*, The 23rd Hiroshima-Sendai Workshop on Number Theory at Sendai., Tohoku University, Japan.

16. June 2024. *"Regularity criterion of mixed characteristic rings via prismatic cohomology"*, Keio Algebra Seminar., Keio University, Japan.

15. June 2024. *"Prisms and regular local rings"*, The 35th Seminar on Commutative Algebra in Japan., Tokushima University, Japan.

14. Apr 2024. *"Prisms and its application to regular rings"*, Saturday Seminar., Meiji University, Japan.

13. Mar 2024. *"Perfectoid ideals and its correspondence"*, The 20th Mathematics Conference for Young Researchers., Hokkaido University, Japan.

12. Feb 2024. *"Mixed characteristic Kunz's theorem with prismatic theory"*, The 28th Conference on Algebra for Young Researchers in Japan., Waseda University, Japan.

11. Dec 2023. *"Commutative ring theoretic approach for the perfectoidization of semiperfectoid rings"*, Number Theory Seminar., Kyoto University, Japan.

10. Nov 2023. *"Ideal correspondence between a perfectoid ring and its tilt"*, The 44th Japan Symposium on Commutative Algebra., LecTore Hayama, Japan.

9. Aug 2023. *"Absolute integral closure"* (Survey talk), The 18th Summer School on Commutative algebra., Tokyo Institute of Technology, Japan.

8. Aug 2023. *"On the relation between perfectoidization and $p$-root closure"*, The 9th China-Japan-Korea International Conference on Ring and Module Theory., Incheon National University, Republic of Korea.

7. July 2023. *"On the commutative ring-theoretic structure of the perfectoidization of semiperfectoid rings"*, The 22nd Hiroshima-Sendai Workshop on Number Theory at Hiroshima., Hiroshima University, Japan.

6. July 2023. *"On the application of perfectoidization to commutative algebra and its structure"*, The 34th Seminar on Commutative Algebra in Japan., Kitami Institute of Technology, Japan.

5. May 2023. *"On Perfectoid(ization) and its commutative ring-theoretic properties"*, Ookayama Youth Seminar in Algebra., Tokyo Institute of Technology, Japan.

4. Mar 2023. *"A mixed characteristic analogue of the perfection of rings"*, The 11th Japan-Vietnam Joint Seminar on Commutative Algebra - by and for young mathematicians -., Vietnam Academy of Science and Technology, Vietnam.

3. Mar 2023. *"An explicit construction of perfectoid almost Cohen-Macaulay algebras in mixed characteristic"*, MSJ Spring Meeting 2023., Chuo University, Japan.

2. Mar 2023. *"An explicit construction of perfectoid almost Cohen-Macaulay algebras"*, The 19th Mathematics Conference for Young Researchers., Hokkaido University, Japan.

1. Nov 2022. *"An explicit construction of perfectoid almost Cohen-Macaulay algebras in mixed characteristic"*, The 43rd Japan Symposium on Commutative Algebra., Osaka University, Japan.

## Membership

- Apr 2023– . Mathematical Society in Japan (MSJ)

## Languages

**Japanese**: Native — *Mother tangue*

**English**: Intermediate — *Can read, write, and, listen but may struggle with conversation*

**French**: Beginner — *Can only read mathematical texts*